

* Most significant risk of oxcarbazepine may be clinically significant hyponatremia (sodium level <125 m mol/L), most likely occurring within the first 3 months of treatment, and occurring in 2–3% of patients

- Unknown if this risk is higher than for carbamazepine

* Since SSRIs can sometimes also reduce sodium due to SIADH (syndrome of inappropriate antidiuretic hormone production), patients treated with combinations of oxcarbazepine and SSRIs should be carefully monitored, especially in the early stages of treatment

- By analogy with carbamazepine, could theoretically be useful in chronic neuropathic pain

Suggested Reading

Beydoun A . Safety and efficacy of oxcarbazepine: results of randomized, double-blind trials . *Pharmacotherapy* 2000 ;**20** (8 Pt 2):152S –158S.

Centorrino F , Albert MJ , Berry JM , et al. Oxcarbazepine: clinical experience with hospitalized psychiatric patients . *Bipolar Disord* 2003 ;**5** :370 –4 .

Dietrich DE , Kropp S , Emrich HM . Oxcarbazepine in affective and schizoaffective disorders . *Pharmacopsychiatry* 2001 ;**34** :242–50 .

Glauser TA . Oxcarbazepine in the treatment of epilepsy . *Pharmacotherapy* 2001 ;**21** :904–19 .

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Vasudev A , Macritchie K , Vasudev K , et al. Oxcarbazepine for acute affective episodes in bipolar disorder . *Cochrane Database Syst Rev* 2011 ;7 (12):CD004857 .

Paliperidone

Therapeutics

Brands

- Invega
- Invega Sustenna
- Invega Trinza
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: dopamine, serotonin receptor antagonist (DS-RAn)
- Atypical antipsychotic (serotonin-dopamine antagonist; second-generation antipsychotics; also a mood stabilizer)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenia (ages 12 and older)**
- **Maintaining response in schizophrenia**

- **Schizoaffective disorder**
- Other psychotic disorders
- Acute mania/mixed mania
- Bipolar maintenance
- Treatment-resistant depression
- Behavioral disturbances in dementia
- Behavioral disturbances in children and adolescents
- Disorders associated with problems with impulse control
- Posttraumatic stress disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis and stabilizing affective symptoms
- Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognitive and affective symptoms
- Interactions at a myriad of other neurotransmitter receptors may contribute to paliperidone's efficacy
- * Serotonin 7 antagonist properties may contribute to antidepressant actions

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior as well as on cognition
- Classically recommended to wait at least 4–6 weeks to determine efficacy of drug, but in practice some patients may require up to 16–20 weeks to show a good response, especially on negative or cognitive symptoms

If It Works

- Most often reduces positive symptoms but does not eliminate them
- Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia
- Most schizophrenia patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Perhaps 5–15% of schizophrenia patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year
- Such patients are considered superresponders or “awakeners” since they may be well enough to be employed, live independently, and sustain long-term relationships
- Continue treatment until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis
- For second and subsequent episodes of psychosis, treatment may need to be indefinite

- Even for first episodes of psychosis, it may be preferable to continue treatment indefinitely to avoid subsequent episodes
- Many bipolar patients may experience a reduction of symptoms by half or more
- Treatment may not only reduce mania but also prevent recurrences of mania in bipolar disorder

If It Doesn't Work

- Try one of the other atypical antipsychotics
- If 2 or more antipsychotic monotherapies do not work, consider clozapine
- Some patients may require treatment with a conventional antipsychotic
- If no first-line atypical antipsychotic is effective, consider higher doses or augmentation with valproate or lamotrigine
- Consider noncompliance and switch to another antipsychotic with fewer side effects
- Consider initiating rehabilitation and psychotherapy such as cognitive remediation
- Consider presence of concomitant drug abuse



Best Augmenting Combos for Partial Response or Treatment Resistance

- Valproic acid (valproate, divalproex, divalproex ER)
- Other mood-stabilizing anticonvulsants (carbamazepine, oxcarbazepine, lamotrigine)
- Topiramate
- Lithium
- Benzodiazepines

Tests

Before starting an atypical antipsychotic

- * Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- * Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile
- Determine if the patient
 - is overweight (BMI 25.0–29.9)
 - is obese (BMI >30)
 - has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
 - has diabetes (fasting plasma glucose >126 mg/dL)
 - has hypertension (BP >140/90 mmHg)

- has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)
- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting an atypical antipsychotic

- * BMI monthly for 3 months, then quarterly
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight
- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic
- * Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness, and clouding of sensorium, even coma

- Should check blood pressure in the elderly before starting and for the first few weeks of treatment
- Monitoring elevated prolactin levels of dubious clinical benefit
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and paliperidone should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin

- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

- * Dose-dependent drug-induced parkinsonism
- * Hyperprolactinemia
- * May increase risk for diabetes and dyslipidemia
- Rare tardive dyskinesia (much reduced risk compared to conventional antipsychotics)
- Sedation, hypersalivation
- Dose-dependent orthostatic hypotension
- Tachycardia
- Injection site reactions
- Tardive dyskinesia (reduced risk compared to conventional antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration



Life-Threatening or Dangerous Side Effects

- Hyperglycemia, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients taking atypical antipsychotics

- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis
- Rare seizures

Weight Gain



- Many experience and/or can be significant in amount
- May be dose-dependent
- May be less than for some antipsychotics, more than for others

Sedation



- Many experience and/or can be significant in amount
- May be dose-dependent
- May be less than for some antipsychotics, more than for others

What to Do About Side Effects

- Wait

- Wait
- Wait
- Anticholinergics may reduce drug-induced parkinsonism when present
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 6 mg/day (oral)
- 39–234 mg/month (Sustenna; see Paliperidone Palmitate section after Pearls for dosing and use)
- 273–819 mg/3 months (Trinza; see Paliperidone Palmitate section after Pearls for dosing and use)

Dosage Forms

- Tablet (extended-release) 1.5 mg, 3 mg, 6 mg, 9 mg
- 1-month injection 39 mg, 78 mg, 117 mg, 156 mg, 234 mg
- 3-month injection 273 mg, 410 mg, 546 mg, 819 mg

How to Dose

- Initial dose 6 mg/day taken in the morning
- Can increase by 3 mg/day every 5 days; maximum dose generally 12 mg/day
- LAI paliperidone is not recommended for patients who have not first demonstrated tolerability to oral paliperidone or risperidone
- See also the Switching section, after Pearls



Dosing Tips –Oral

- Tablet should not be divided or chewed, but rather should only be swallowed whole

- Tablet does not change shape in the gastrointestinal tract and generally should not be used in patients with gastrointestinal narrowing because of the risk of intestinal obstruction
- Some patients may benefit from doses above 6 mg/day; alternatively, for some patients 3 mg/day may be sufficient
- A common dosage error is to assume the paliperidone ER oral dose is the same as the risperidone oral dose in mg, and that paliperidone should be titrated. However, many patients do well initiating a dose of 6 mg orally of paliperidone ER without titration
- There is a dose-dependent increase in some side effects, including drug-induced parkinsonism and weight gain, above 6 mg/day
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- For treatment resistance, consider obtaining plasma drug levels to guide dosing above recommended levels
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

- Oral doses correspond to injection doses as follows: 3 mg oral to 39–78 mg injection, 6 mg oral to 117 mg injection, 12 mg oral to 234 mg injection

Overdose

- Drug-induced parkinsonism, gait unsteadiness, sedation, tachycardia, hypotension, QT prolongation

Long-Term Use

- Approved for maintenance in schizophrenia
- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- See Switching section of individual agents for how to stop paliperidone ER and LAI
- Rapid oral discontinuation may lead to rebound psychosis and worsening of symptoms

Pharmacokinetics

- Active metabolite of risperidone

- Half-life approximately 23 hours
- Absorption is reduced if taken on an empty stomach

Drug Interactions

- May increase effects of antihypertensive agents
- May antagonize levodopa, dopamine agonists
- May enhance QTc prolongation of other drugs capable of prolonging QTc interval

Other Warnings/Precautions

- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Dysphagia has been associated with antipsychotic use, and paliperidone should be used cautiously in patients at risk for aspiration pneumonia
- Paliperidone prolongs QTc interval more than some other antipsychotics

- Priapism has been reported with other antipsychotics, including risperidone
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient is taking agents capable of significantly prolonging QTc interval (e.g., pimozide, thioridazine, selected antiarrhythmics, moxifloxacin, sparfloxacin)
- If there is a history of QTc prolongation or cardiac arrhythmia, recent acute myocardial infarction, uncompensated heart failure
- If patient has a preexisting severe gastrointestinal narrowing
- If there is a proven allergy to paliperidone or risperidone

Special Populations

Renal Impairment

- For mild impairment, maximum recommended dose 6 mg/day

- For moderate impairment, initial and maximum recommended dose 3 mg/day
- For severe impairment, initial dose 1.5 mg/day; maximum recommended dose 3 mg/day

Hepatic Impairment

- No dose adjustment necessary for mild to moderate impairment
- Use in individuals with severe hepatic impairment has not been studied

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension
- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist

Elderly

- Some patients may tolerate lower doses better
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to

placebo, and also have an increased risk of cerebrovascular events

- Consider pimavanserin for dementia-related psychosis or Parkinson's disease psychosis instead of an atypical antipsychotic

Children and Adolescents

- Safety and efficacy have not been established under age 12
- Adolescents <51 kg: initial 3 mg/day; recommended 3–6 mg/day; maximum 6 mg/day
- Adolescents >51 kg: initial 3 mg/day; recommended 3–12 mg/day; maximum 12 mg/day
- Children and adolescents using paliperidone may need to be monitored more often than adults and may tolerate lower doses better

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during

the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding

- When administered to rats during the period of organogenesis, there were no treatment-related effects at doses up to 10 times the maximum recommended human dose (MRHD)
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Paliperidone may be preferable to anticonvulsant mood stabilizers if treatment is required during pregnancy
- Effects of hyperprolactinemia on the fetus are unknown
- National Pregnancy Registry for Atypical Antipsychotics: 1-866-961-2388,
<https://womensmentalhealth.org/research/pregnancyregistry/atypical/antipsychotic>

Breast Feeding

- Some drug is found in mother's breast milk
- Recommended either to discontinue drug or bottle feed
- Infants of women who choose to breast feed while on paliperidone should be monitored for possible adverse effects; sedation, failure to thrive, jitteriness, and drug-induced parkinsonism (tremor and abnormal muscle movements) have been reported

The Art of Psychopharmacology

Potential Advantages

- Some cases of psychosis and bipolar disorder refractory to treatment with other antipsychotics
- Patients requiring rapid onset of antipsychotic action without dosage titration

Potential Disadvantages

- Patients for whom elevated prolactin may not be desired (e.g., possibly pregnant patients; pubescent girls with amenorrhea; postmenopausal women with low estrogen who do not take estrogen replacement therapy)

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive symptoms
- Unstable mood (both depression and mania)
- Aggressive symptoms



Pearls

- Some patients respond to paliperidone or tolerate paliperidone better than the parent drug risperidone

- Hyperprolactinemia in women with low estrogen may accelerate osteoporosis
- Less weight gain than some antipsychotics, more than others
- May cause more motor side effects than some other atypical antipsychotics, especially when administered to patients with Parkinson's disease or Lewy body dementia
- Trilayer tablet consists of 3 compartments (2 containing drug, 1 a "push" compartment) and an orifice at the head of the first drug compartment; water fills the push compartment and gradually pushes drug up and out of the tablet through the orifice
- LAI does not require simultaneous oral medication
- LAI may work better and better after a few weeks of treatment in some patients
- LAI may be very well tolerated
- LAI may be combined with a second antipsychotic administered orally for difficult cases
- A 6-month long-acting injectable formulation is in development
- Patients with inadequate responses to atypical antipsychotics may benefit from determination of plasma drug levels and, if low, a dosage increase even beyond the usual prescribing limits if tolerated; minimum therapeutic level is 28 ng/mL with point of futility of 112 ng/mL if effective and tolerated

- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring
- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic
- Patients with inadequate responses to atypical antipsychotics may also benefit from a trial of augmentation with a conventional antipsychotic or switching to a conventional antipsychotic
- However, long-term polypharmacy with a combination of a conventional antipsychotic with an atypical antipsychotic may combine their side effects without clearly augmenting the efficacy of either
- Although a frequent practice by some prescribers, adding 2 conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy

Paliperidone Palmitate

	1-month (Sustenna)	3-month (Trinza)
Vehicle	Water	Water
Tmax	13 days	30–33 days

	1-month (Sustenna)	3-month (Trinza)
T1/2 with multiple dosing	25–49 days	84–95 days (deltoid) 118–139 days (gluteal)
Time to reach steady state	1 week	
Able to be loaded	Yes	N/A
Dosing schedule (maintenance)	4 weeks	12 weeks
Injection site	Intramuscular (deltoid at initiation, then either deltoid or gluteal)	Intramuscular (deltoid or gluteal)
Needle gauge	22 or 23	22
Dosage forms	39 mg, 78 mg, 117 mg, 156 mg, 234 mg	273 mg, 410 mg, 546 mg, 819 mg
Injection volume	156 mg/mL; range 0.25–1.5 mL	0.875–2.625 mL

Usual Dosage Range

- 1-month injectable maintenance dose: 117 mg/month (range 39–234 mg/month)

- 3-month injectable maintenance dose: 273–819 mg/3 months

How to Dose –1-month Injectable

- Not recommended for patients who have not first demonstrated tolerability to oral paliperidone or risperidone (in clinical trials, 2 oral or short-acting IM doses are generally used to establish tolerability)
- Conversion from oral: 234 mg delivered intramuscularly in the deltoid on day 1; 156 mg delivered intramuscularly in the deltoid on day 8; maintenance dose should start 4 weeks after the 2nd loading injection

Oral Equivalence (Approximate)	
Oral paliperidone	1-month Sustenna
3 mg	39–78 mg
6 mg	117 mg
9 mg	156 mg
12 mg	234 mg

How to Dose –3-month Injectable

- Only for patients who have received adequate treatment with paliperidone Sustenna for at least 4 months